

SOHCAHTOA

Higher Tier

Right-angled trigonometry and applications

Name

Date

35 questions

126 marks

2017-2025

Instructions

- Answer each question in the space provided.
- Show all your working and follow the rounding instructions.
- Diagrams are not accurately drawn unless stated otherwise.
- Use a calculator except for pages marked NON-CALCULATOR.

Using this booklet

Questions are arranged by exam series. Each page gives its original paper reference and question number. The question numbers within the extracted artwork are the original exam numbers.

Includes multi-step questions involving 3D shapes, bounds, algebra and other geometry skills.

This is an independently compiled practice resource, not an official Pearson examination paper. The selected questions run through June 2025; November 2025 is not included.

Question index

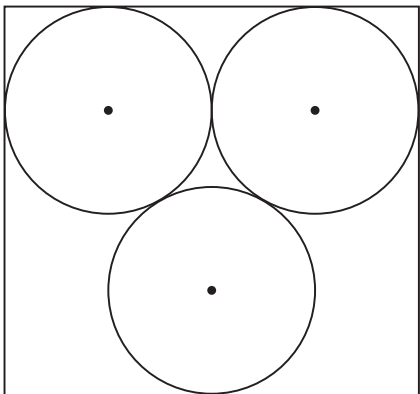
Higher Tier | 35 questions | 126 marks

Practice	Exam series	Paper	Original question	Marks	Page
01	June 2017	1MA1/2H	Q21	4	3
02	June 2017	1MA1/3H	Q7	2	4
03	June 2017	1MA1/3H	Q18	5	5
04	November 2017	1MA1/2H	Q7	5	6
05	November 2017	1MA1/2H	Q20	5	7
06	June 2018	1MA1/2H	Q18	4	8
07	June 2018	1MA1/3H	Q5	3	9
08	June 2018	1MA1/3H	Q19	5	10
09	November 2018	1MA1/1H	Q8(b)	2	11
10	November 2018	1MA1/3H	Q6	4	12
11	November 2018	1MA1/3H	Q12	4	13
12	June 2019	1MA1/2H	Q5	2	14
13	June 2019	1MA1/2H	Q19	4	15
14	June 2019	1MA1/3H	Q14	4	16
15	June 2019	1MA1/3H	Q22	4	17
16	November 2019	1MA1/2H	Q12	4	18
17	November 2020	1MA1/2H	Q5	2	19
18	November 2020	1MA1/3H	Q9	5	20
19	November 2021	1MA1/1H	Q18	5	21
20	November 2021	1MA1/2H	Q6	4	22
21	June 2022	1MA1/3H	Q12	3	23
22	June 2022	1MA1/3H	Q18	5	24
23	November 2022	1MA1/2H	Q5	2	25
24	November 2022	1MA1/2H	Q20	4	26
25	November 2022	1MA1/3H	Q26	5	27
26	June 2023	1MA1/1H	Q22	2	28
27	June 2023	1MA1/2H	Q19	3	29
28	June 2023	1MA1/3H	Q10	3	30
29	November 2023	1MA1/3H	Q24	4	31
30	June 2024	1MA1/2H	Q10	3	32
31	June 2024	1MA1/3H	Q21	4	33
32	November 2024	1MA1/2H	Q20	3	34
33	November 2024	1MA1/3H	Q8	2	35
34	June 2025	1MA1/2H	Q7	2	36
35	June 2025	1MA1/2H	Q19	4	37

The original exam numbering is retained in each question.

Paper 1H questions are marked NON-CALCULATOR on their pages.

21 The diagram shows 3 identical circles inside a rectangle.
Each circle touches the other two circles and the sides of the rectangle, as shown in the diagram.



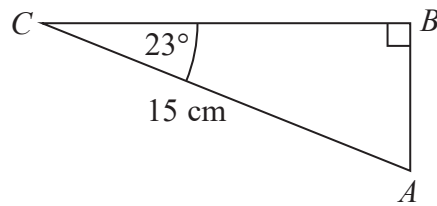
The radius of each circle is 24 mm.

Work out the area of the rectangle.

Give your answer correct to 3 significant figures.

..... mm²

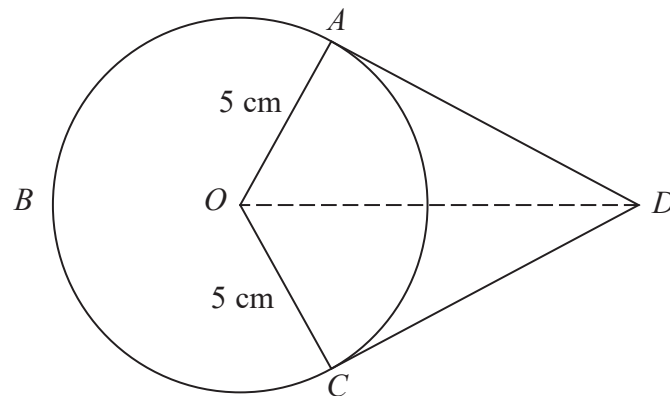
(Total for Question 21 is 4 marks)



.....cm

(Total for Question 7 is 2 marks)

18



A , B and C are points on a circle of radius 5 cm, centre O .

DA and DC are tangents to the circle.

$DO = 9$ cm

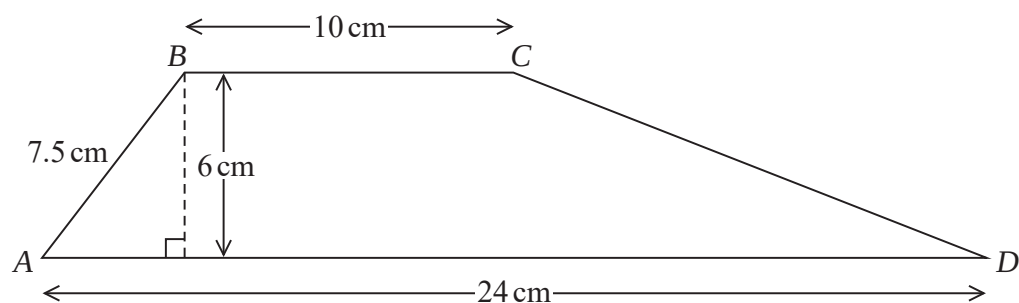
Work out the length of arc ABC .

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 18 is 5 marks)

7 $ABCD$ is a trapezium.

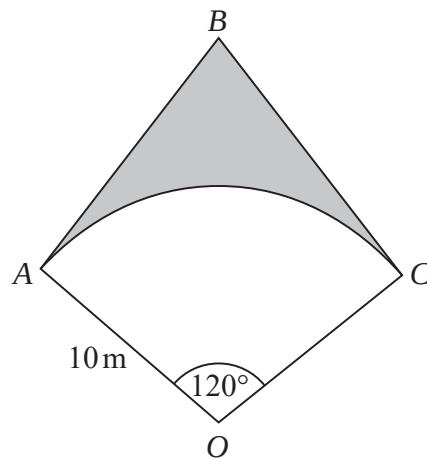


Work out the size of angle CDA .

Give your answer correct to 1 decimal place.

(Total for Question 7 is 5 marks)

20



OAC is a sector of a circle, centre O , radius 10 m .

BA is the tangent to the circle at point A .

BC is the tangent to the circle at point C .

Angle $AOC = 120^\circ$

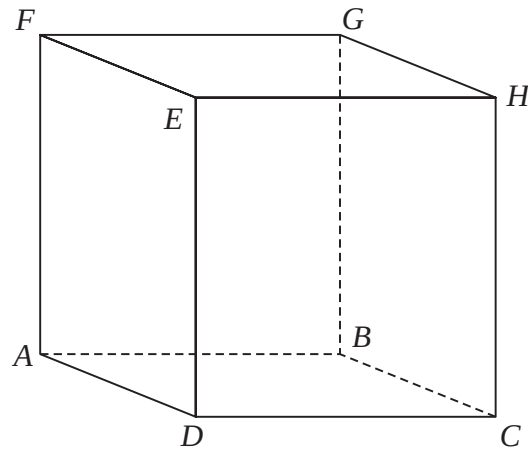
Calculate the area of the shaded region.

Give your answer correct to 3 significant figures.

..... m^2

(Total for Question 20 is 5 marks)

18 $ABCDEFGH$ is a cuboid.



$$AB = 7.3 \text{ cm}$$

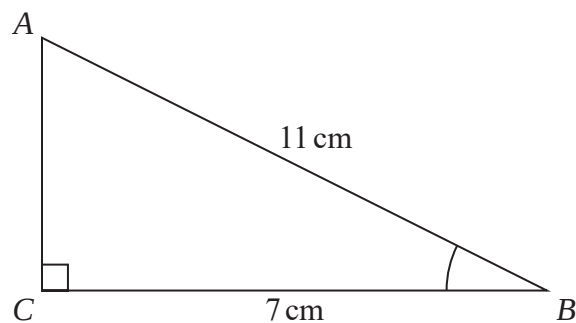
$$CH = 8.1 \text{ cm}$$

$$\text{Angle } BCA = 48^\circ$$

Find the size of the angle between AH and the plane $ABCD$.
Give your answer correct to 1 decimal place.

(Total for Question 18 is 4 marks)

5 ABC is a right-angled triangle.



- (a) Work out the size of angle ABC .
Give your answer correct to 1 decimal place.

.....
(2)

The length of the side AB is reduced by 1 cm.

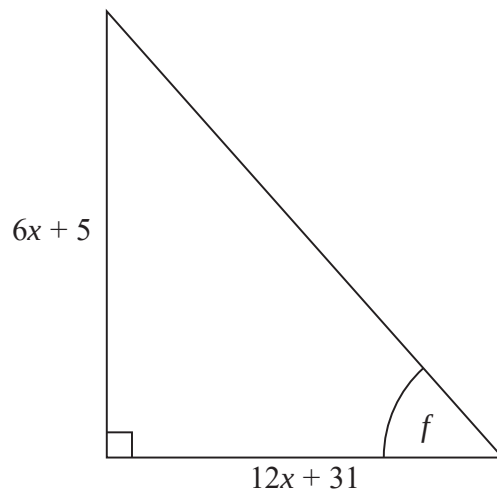
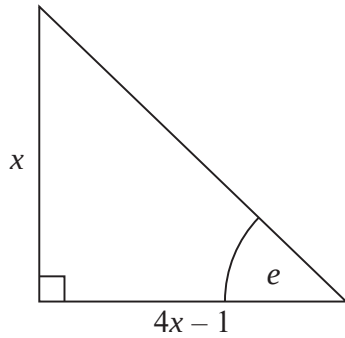
The length of the side BC is still 7 cm.
Angle ACB is still 90°

- (b) Will the value of $\cos ABC$ increase or decrease?
You must give a reason for your answer.

.....
(1)

(Total for Question 5 is 3 marks)

19 Here are two right-angled triangles.



Given that

$$\tan e = \tan f$$

find the value of x .

You must show all your working.

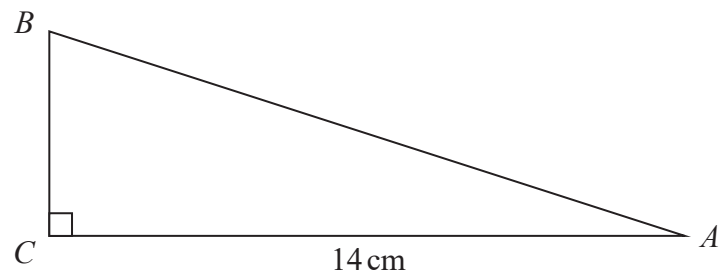
(Total for Question 19 is 5 marks)

A right-angled triangle is shown. The vertical side is labeled 4 cm. The hypotenuse is labeled x cm. The angle between the vertical side and the hypotenuse is labeled 60° . A right-angle symbol is at the bottom-left vertex.

(b) Work out the value of x .

Additional working space

6 ABC is a right-angled triangle.



$AC = 14\text{ cm}$.
Angle $C = 90^\circ$

size of angle B : size of angle $A = 3 : 2$

Work out the length of AB .
Give your answer correct to 3 significant figures.

.....cm

(Total for Question 6 is 4 marks)

Additional working space

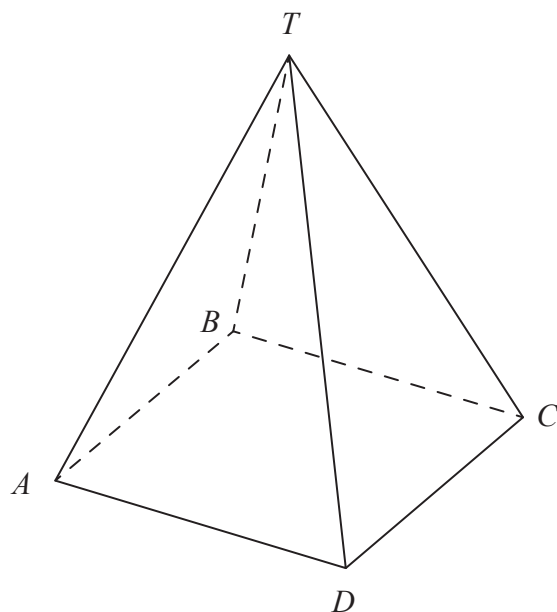
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12 Here is a pyramid with a square base $ABCD$.



$$AB = 5 \text{ m}$$

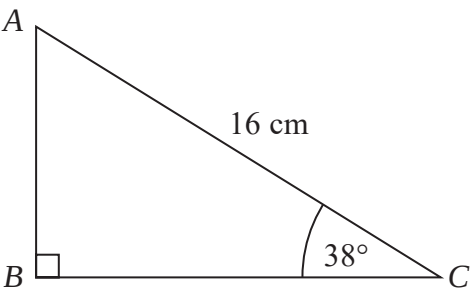
The vertex T is 12 m vertically above the midpoint of AC .

Calculate the size of angle TAC .

○

(Total for Question 12 is 4 marks)

5 ABC is a right-angled triangle.



Calculate the length of AB .
Give your answer correct to 2 decimal places.

.....cm

(Total for Question 5 is 2 marks)

Additional working space

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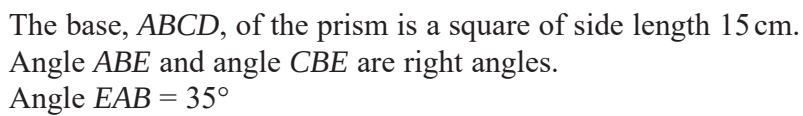
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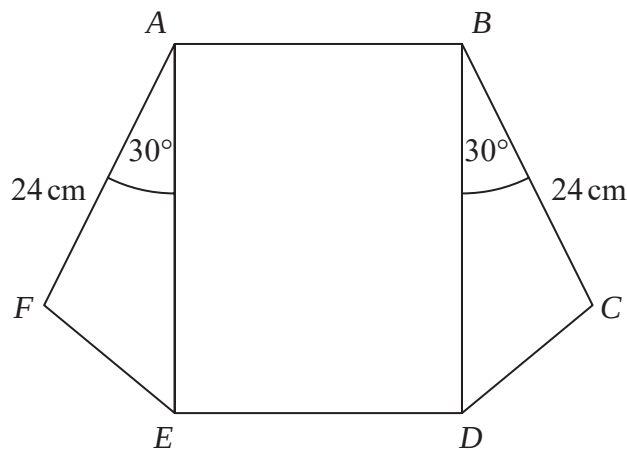
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$$DM:MA = 2:3$$

(Total for Question 19 is 4 marks)

14 The diagram shows a rectangle, $ABDE$, and two congruent triangles, AFE and BCD .



area of rectangle $ABDE$ = area of triangle AFE + area of triangle BCD

$AB : AE = 1 : 3$

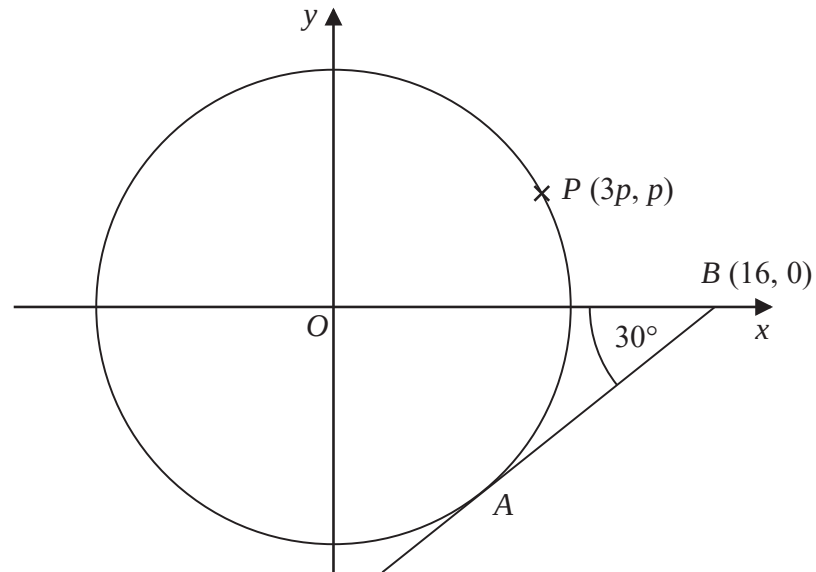
Work out the length of AE .

..... cm

(Total for Question 14 is 4 marks)

Additional working space

22 The diagram shows a circle, centre O .



AB is the tangent to the circle at the point A .
Angle $OBA = 30^\circ$

Point B has coordinates $(16, 0)$

Point P has coordinates $(3p, p)$

Find the value of p .

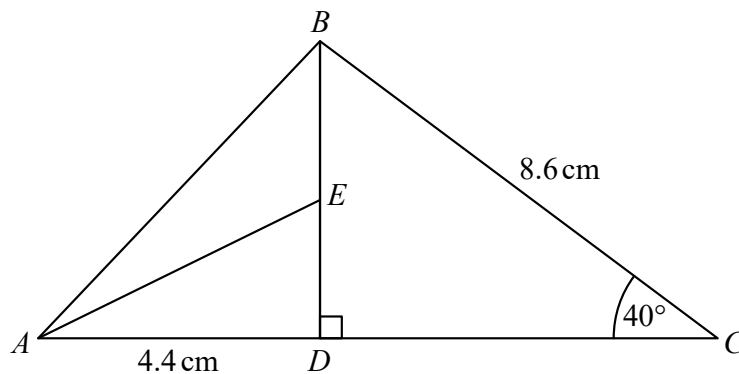
Give your answer correct to 1 decimal place.

You must show all your working.

$p = \dots\dots\dots$

(Total for Question 22 is 4 marks)

12 The diagram shows triangle ABC .



ADC and DEB are straight lines.

$AD = 4.4 \text{ cm}$

$BC = 8.6 \text{ cm}$

E is the midpoint of DB .

Angle $CDB = 90^\circ$

Angle $DCB = 40^\circ$

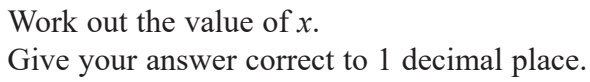
Work out the size of angle EAD .

Give your answer correct to 1 decimal place.

You must show all your working.

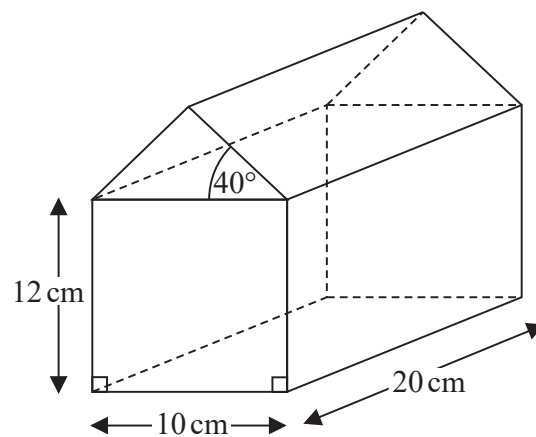
○

(Total for Question 12 is 4 marks)



Additional working space

- 9 The diagram shows a prism.



The cross section of the prism has exactly one line of symmetry.

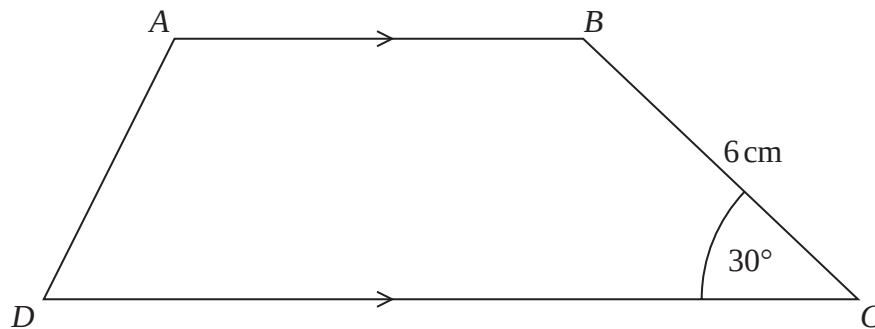
Work out the volume of the prism.

Give your answer correct to 3 significant figures.

..... cm³

(Total for Question 9 is 5 marks)

18 Here is trapezium $ABCD$.



The area of the trapezium is 66 cm^2

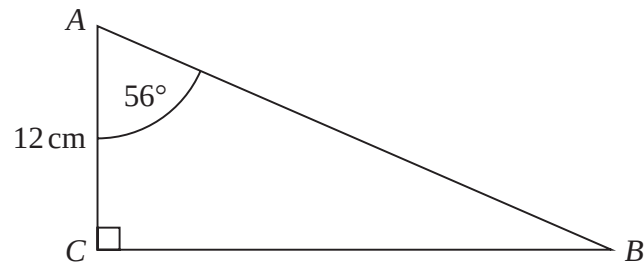
the length of AB : the length of $CD = 2:3$

Find the length of AB .

..... cm

(Total for Question 18 is 5 marks)

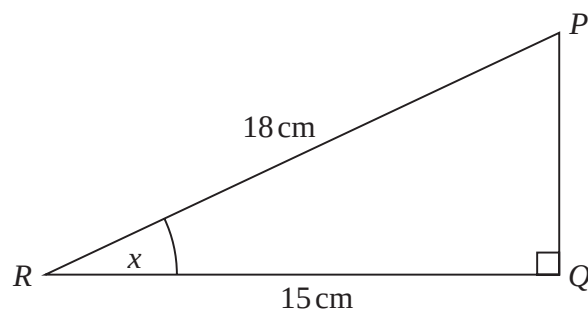
6 ABC is a right-angled triangle.



- (a) Work out the length of BC .
Give your answer correct to 1 decimal place.

..... cm
(2)

PQR is a right-angled triangle.

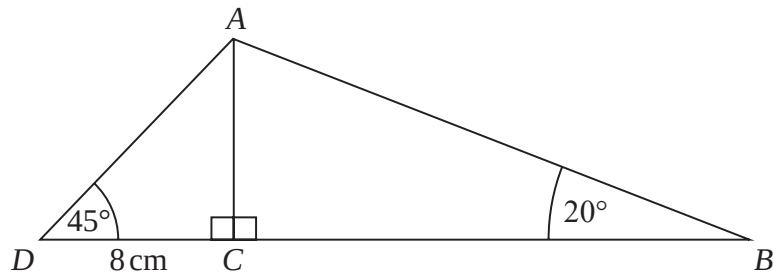


- (b) Work out the size of the angle marked x .
Give your answer correct to 1 decimal place.

.....
(2)

(Total for Question 6 is 4 marks)

12 ABC and ACD are right-angled triangles.



$DC = 8\text{ cm}$

Angle $ADC = 45^\circ$

Angle $ABC = 20^\circ$

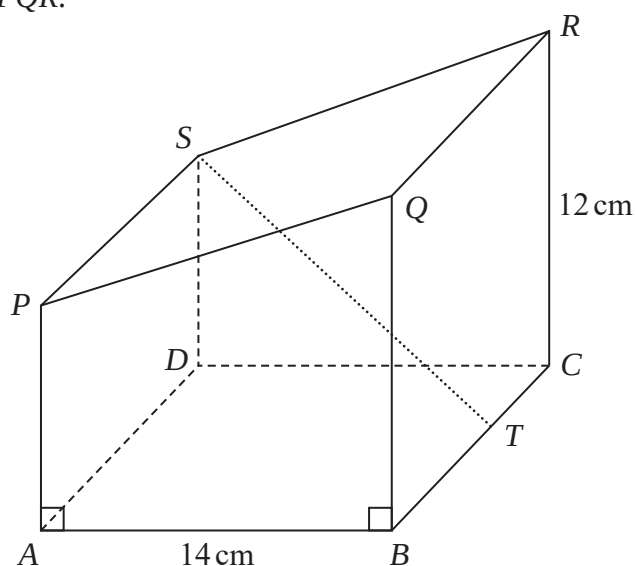
Work out the length of AB .

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 12 is 3 marks)

18 Here is a prism $ABCDSPQR$.



The base $ABCD$ of the prism is a square of side 14 cm

T is the point on BC such that $BT : TC = 4 : 3$

The cross section of the prism is in the shape of a trapezium of area 147 cm^2

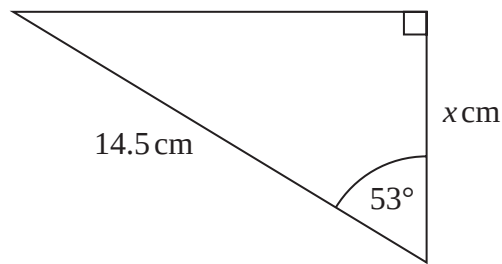
$CR = 12\text{ cm}$

Find the size of the angle between the line ST and the base $ABCD$.

Give your answer correct to 1 decimal place.

(Total for Question 18 is 5 marks)

5



Work out the value of x .
Give your answer correct to 3 significant figures.

 $x = \dots\dots\dots$

(Total for Question 5 is 2 marks)

Additional working space

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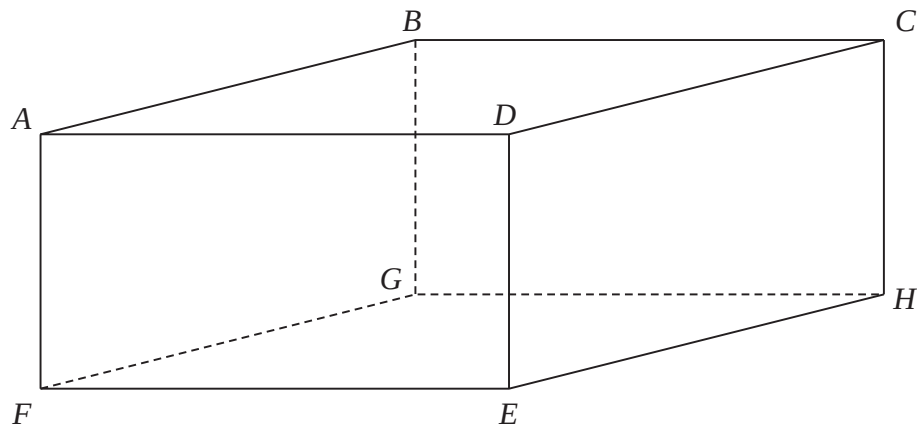
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20 $ABCDEFGH$ is a cuboid.



$$AD = 9 \text{ cm}$$

$$FD = 13 \text{ cm}$$

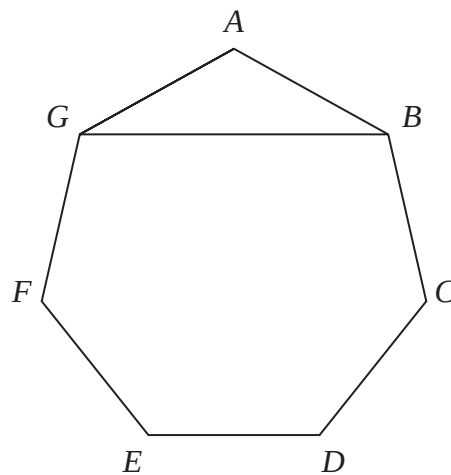
$$\text{Angle } GHF = 49^\circ$$

Work out the size of angle FAH .

Give your answer correct to the nearest degree.

(Total for Question 20 is 4 marks)

26 $ABCDEFG$ is a regular heptagon.



The area of triangle ABG is 30 cm^2

Calculate the length of GB .

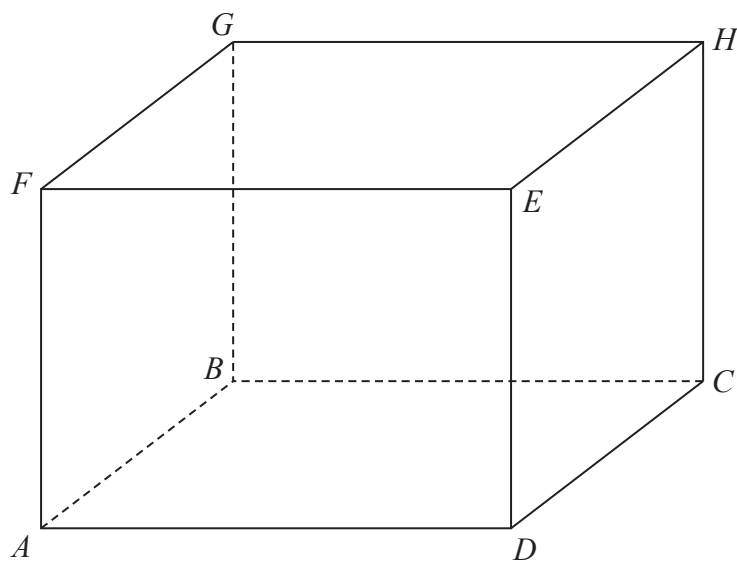
Give your answer correct to 3 significant figures.

You must show all your working.

..... cm

(Total for Question 26 is 5 marks)

22 $ABCDEFGH$ is a cuboid.



$$AF = 6.8 \text{ cm}$$

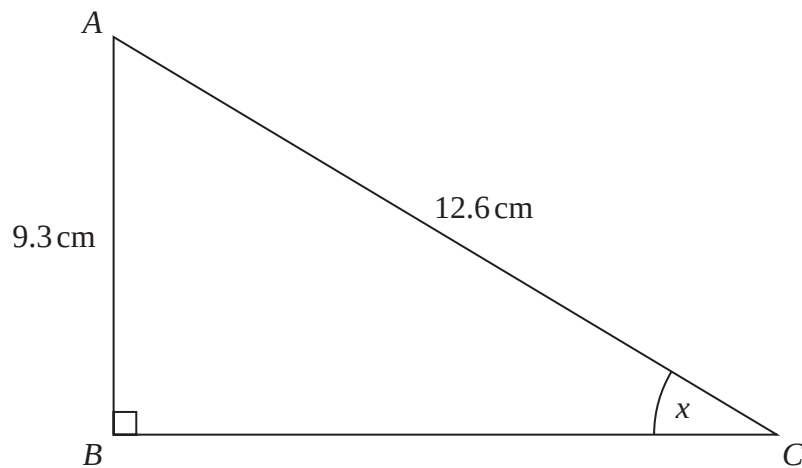
$$FC = 13.6 \text{ cm}$$

Work out the size of the angle between FC and the plane $ABCD$.

○

(Total for Question 22 is 2 marks)

19 ABC is a right-angled triangle.



$AB = 9.3\text{ cm}$ correct to the nearest mm.

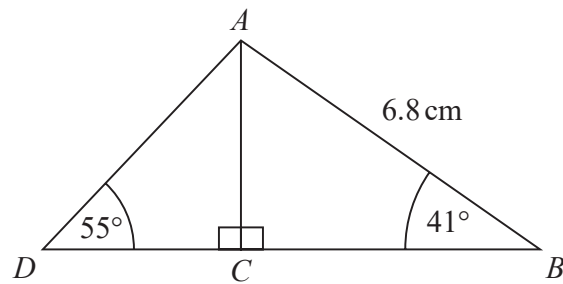
$AC = 12.6\text{ cm}$ correct to the nearest mm.

Calculate the lower bound for the size of the angle marked x .

You must show all your working.

.....
(Total for Question 19 is 3 marks)

- 10 ABD is a triangle.
 C is a point on BD .

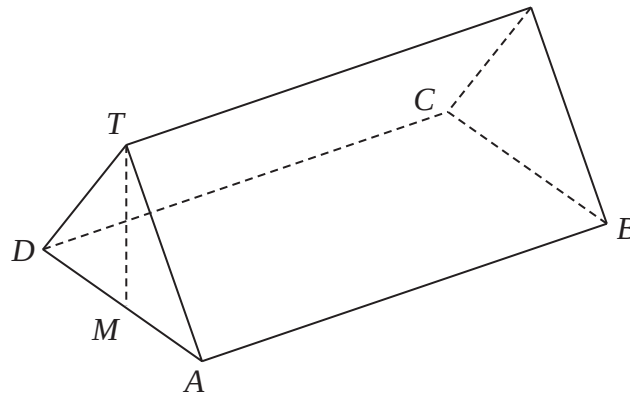


Work out the length of DC .
Give your answer correct to 1 decimal place.

..... cm

(Total for Question 10 is 3 marks)

24 The diagram shows a triangular prism with a horizontal rectangular base $ABCD$.



M is the midpoint of AD .

The vertex T of the prism is vertically above M .

$$AB = 14.7 \text{ cm}$$

$$BC = 3.8 \text{ cm}$$

$$MT = 2.3 \text{ cm}$$

P is the point on AB such that

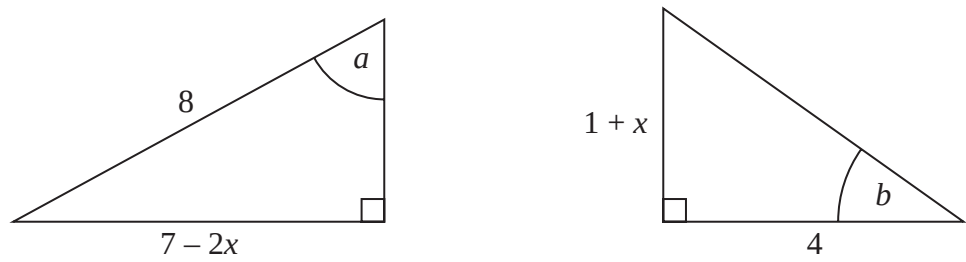
$$AP:PB = 5:2$$

Calculate the size of the angle between TP and the base $ABCD$ of the prism.
Give your answer correct to 1 decimal place.

o

(Total for Question 24 is 4 marks)

10 The diagram shows two right-angled triangles.



All lengths are measured in centimetres.
Given that

$\sin a = \tan b$
work out the value of x .

$x = \dots\dots\dots$

(Total for Question 10 is 3 marks)

Additional working space

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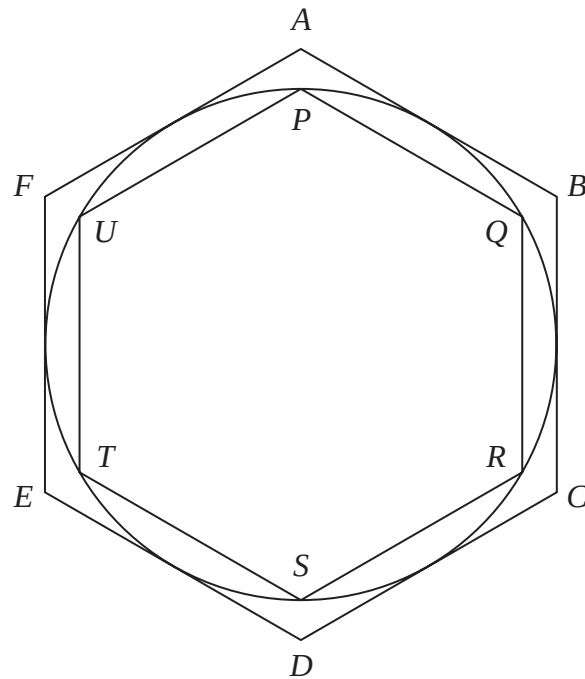
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21 The diagram shows a circle, radius r cm and two regular hexagons.



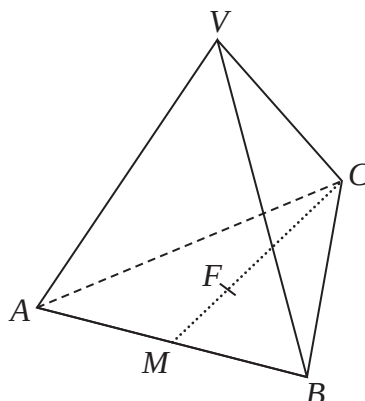
Each side of the larger hexagon $ABCDEF$ is a tangent to the circle.
Each side of the smaller hexagon $PQRSTU$ is a chord of the circle.

By considering perimeters, show that

$$3 < \pi < 2\sqrt{3}$$

(Total for Question 21 is 4 marks)

- 20 $VABC$ is a solid pyramid.
 ABC is an equilateral triangle.



M is the midpoint of AB .
 F is the point on MC such that $MF:FC = 1:2$

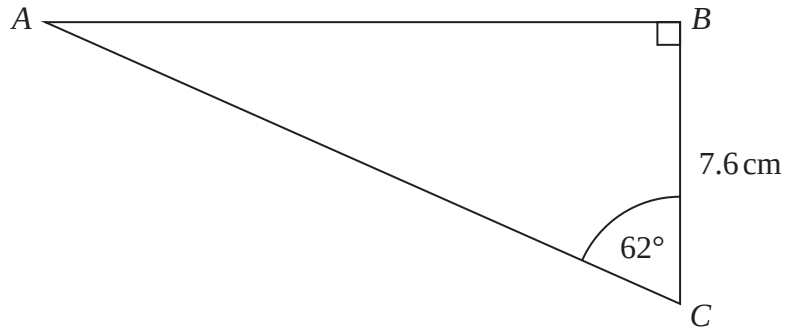
The vertex V is vertically above F .
 $VA = VB = VC$

$VF = 8 \text{ cm}$ Angle $VCM = 52^\circ$

Work out the side length of the equilateral triangle ABC .
 Give your answer correct to 1 decimal place.

..... cm

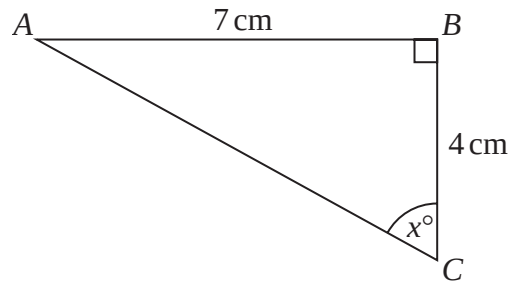
(Total for Question 20 is 3 marks)



..... cm

(Total for Question 8 is 2 marks)

7 ABC is a right-angled triangle.



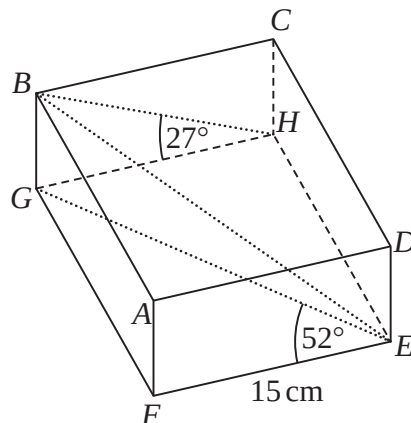
Calculate the value of x .
Give your answer correct to 1 decimal place.

$x =$

(Total for Question 7 is 2 marks)

Additional working space

19 $ABCDEFGH$ is a cuboid.



Angle $GEF = 52^\circ$

Angle $BHG = 27^\circ$

$EF = 15 \text{ cm}$

Work out the size of angle GEB .

Give your answer to the nearest degree.

(Total for Question 19 is 4 marks)